

Maxwell Relations in Thermodynamics: Theoretical Foundations and a Computational Approach

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Abstract

This project presents a theoretical and computational development of the Maxwell relations in thermodynamics. Starting from thermodynamic potentials and the concept of exact differentials, we show why the Maxwell relations are general identities, independent of the specific material under consideration. These relations are then used to connect experimentally accessible derivatives with otherwise difficult-to-measure quantities. The theoretical structure is subsequently translated into a computational framework in Python to calculate and visualize entropy, enthalpy, internal energy, and free energies for ideal and non-ideal thermodynamic models.

1 Problem Statement

In thermodynamics, macroscopic properties of a system are described by state variables such as pressure P , temperature T , volume V , and entropy S . In practice, not all partial derivatives of these variables can be measured or computed directly. For example, determining how entropy changes with volume at constant temperature is generally non-trivial.

However, thermodynamic theory imposes a strong mathematical structure: thermodynamic potentials are state functions, and their differentials are exact. The existence of exact differentials implies symmetry relations between mixed second-order partial derivatives. These symmetries lead directly to the Maxwell relations, which connect seemingly unrelated derivatives and allow one to express difficult quantities (such as entropy derivatives) in terms of more accessible ones (such as pressure or temperature derivatives).

The central goal of this project is to address the following questions:

1. Why do the Maxwell relations hold from first principles?
2. How can these relations be used in practice to compute thermodynamic properties within a computational framework?

2 General Objective

To develop and validate a theoretical–computational framework that derives the Maxwell relations from thermodynamic potentials and applies them to the calculation and visualization of entropy, enthalpy, internal energy, and free energies in both ideal and non-ideal thermodynamic systems.

3 Specific Objectives

1. Formulate the thermodynamic potentials and identify their natural variables.
2. Formally derive the four Maxwell relations using exact differentials and mixed partial derivatives.
3. Provide a physical interpretation of each Maxwell relation and its practical relevance.
4. Implement a Python-based computational pipeline to evaluate partial derivatives, integrate relations, and visualize thermodynamic quantities.
5. Compare ideal systems (ideal gas) with non-ideal models (e.g., Van der Waals equation of state).

4 Scope and Assumptions

This work is developed within the framework of classical equilibrium thermodynamics, under the following assumptions:

- Macroscopic systems in thermodynamic equilibrium or quasi-equilibrium.
- Well-defined state variables and differentiable state functions.
- Time-dependent and non-equilibrium effects are neglected in the initial stage.

The computational scope is initially limited to simple compressible systems and can be extended to include additional variables such as chemical potential μ and particle number N .

5 Notation

Standard partial-derivative notation is used throughout. For example,

$$\left(\frac{\partial S}{\partial V}\right)_T$$

denotes the change of entropy with respect to volume at constant temperature. The following thermodynamic potentials, expressed in their natural variables, are employed:

$$U(S, V), \quad H(S, P), \quad F(T, V), \quad G(T, P).$$